OSITO DRONE STEM ACADEMY

SERIES 01: COLOR-BY-NUMBER DRONE FLEET - AGES 4-6 (PRE-K & K)

THIS STEM COLORING BOOK BELONGS TO

YOUNG PILOT'S NAME / NOMBRE DEL PILOTO

MY DRONE'S CALLSIGN / NOMBRE DE MI DRON

CO-PILOT / GROWN-UP CO-PILOT NAME

Title: Vol. 001: Maya's Count the 4 Propellers at Organic Sunflower & Corn Farm

Featured STEM Pilot: Maya (Young STEM Pilot with Afro Puffs & Lab Goggles)

Flight Environment: Organic Sunflower & Corn Farm - Target Age: Ages 4-6 (Pre-K & K)

Core STEM & Math Focus: Counting by 4s & Rotor Symmetry

Includes: 8 Single-Sided Coloring Plates, 4 Vector STEM Activities, Flight Log & Certificate!

Explore Youth Drone STEM Camps, 3D Printing & Flight Simulators at ositodrone.com

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OFFICIAL YOUTH DRONE SAFETY RULES & FAA TRUST NOTICE

SAFETY RULES FOR RECREATIONAL DRONE PILOTS

1

1. ALWAYS FLY WITH A GROWN-UP

Make sure a parent, teacher, or coach supervises every flight.

2

2. FLY IN OPEN, SAFE SPACES

Stay clear of people, pets, roads, trees, and active airport airspace.

3

3. KEEP DIRECT LINE-OF-SIGHT

Always keep your quadcopter where you can see it with your own eyes.

4

4. USE DUCTED PROPELLER GUARDS

Inspect all 4 ducted prop guards and battery latch before takeoff.

5

5. COUNT & CHECK PRE-FLIGHT MATH

4 propellers + 1 charged battery + 1 adult co-pilot = Safe Takeoff!

6

6. TAKE THE FREE FAA TRUST TEST

Learn official recreational sky rules at faa.gov/uas/trust with family.

OSITO DRONE LLC – PUBLISHING & LEGAL ATTRIBUTION

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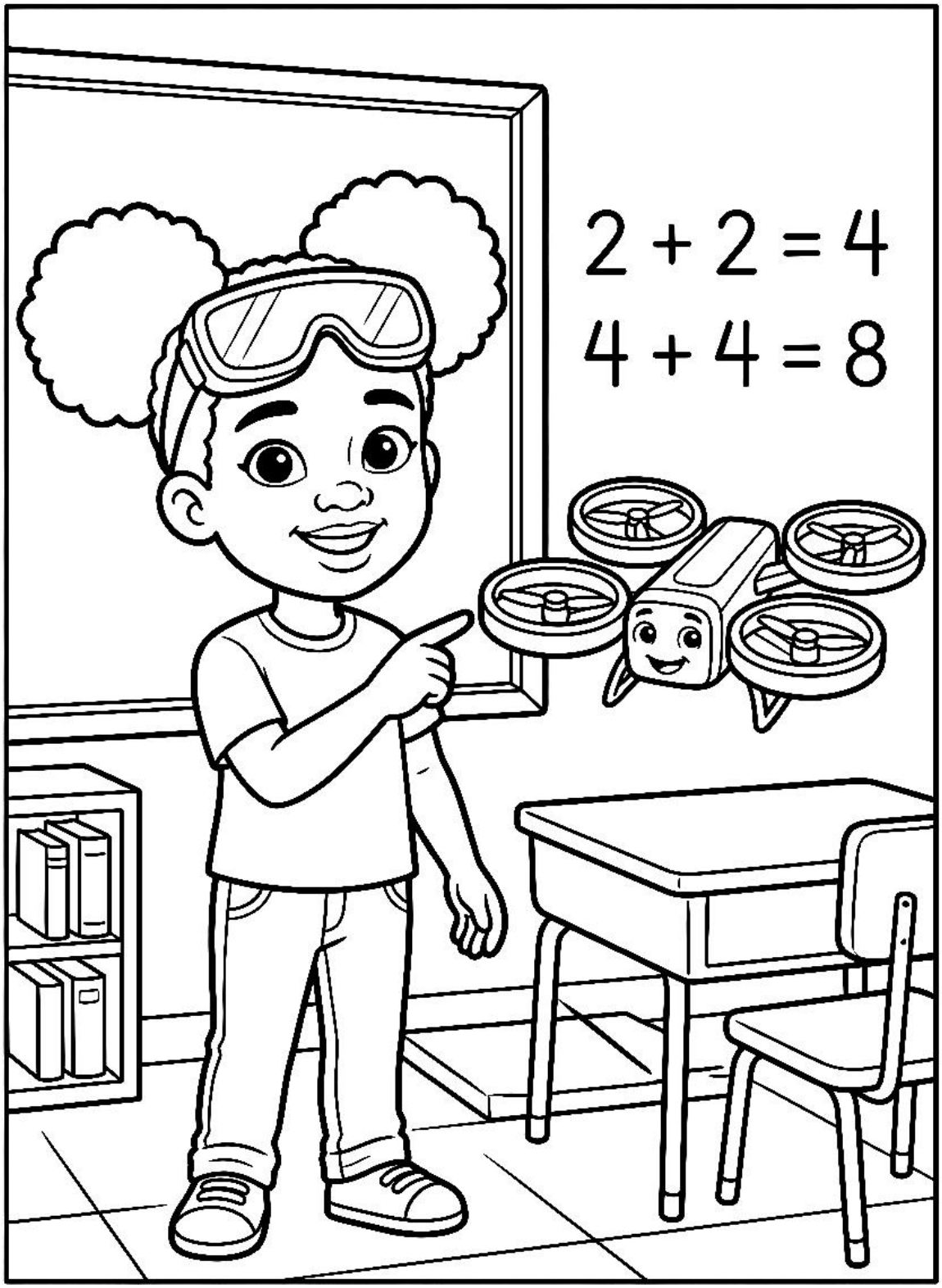
Publisher Contact: legal@ositodrone.com - Website: ositodrone.com

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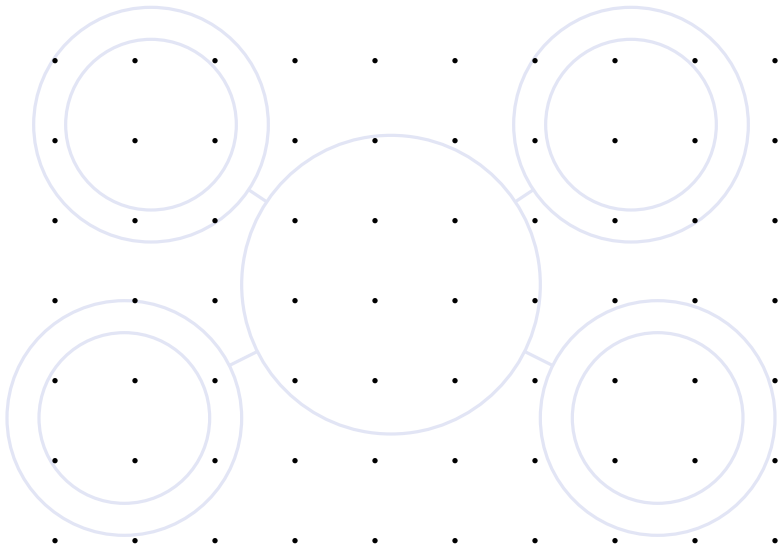
Designed to comply with FAA TRUST youth aviation educational benchmarks.

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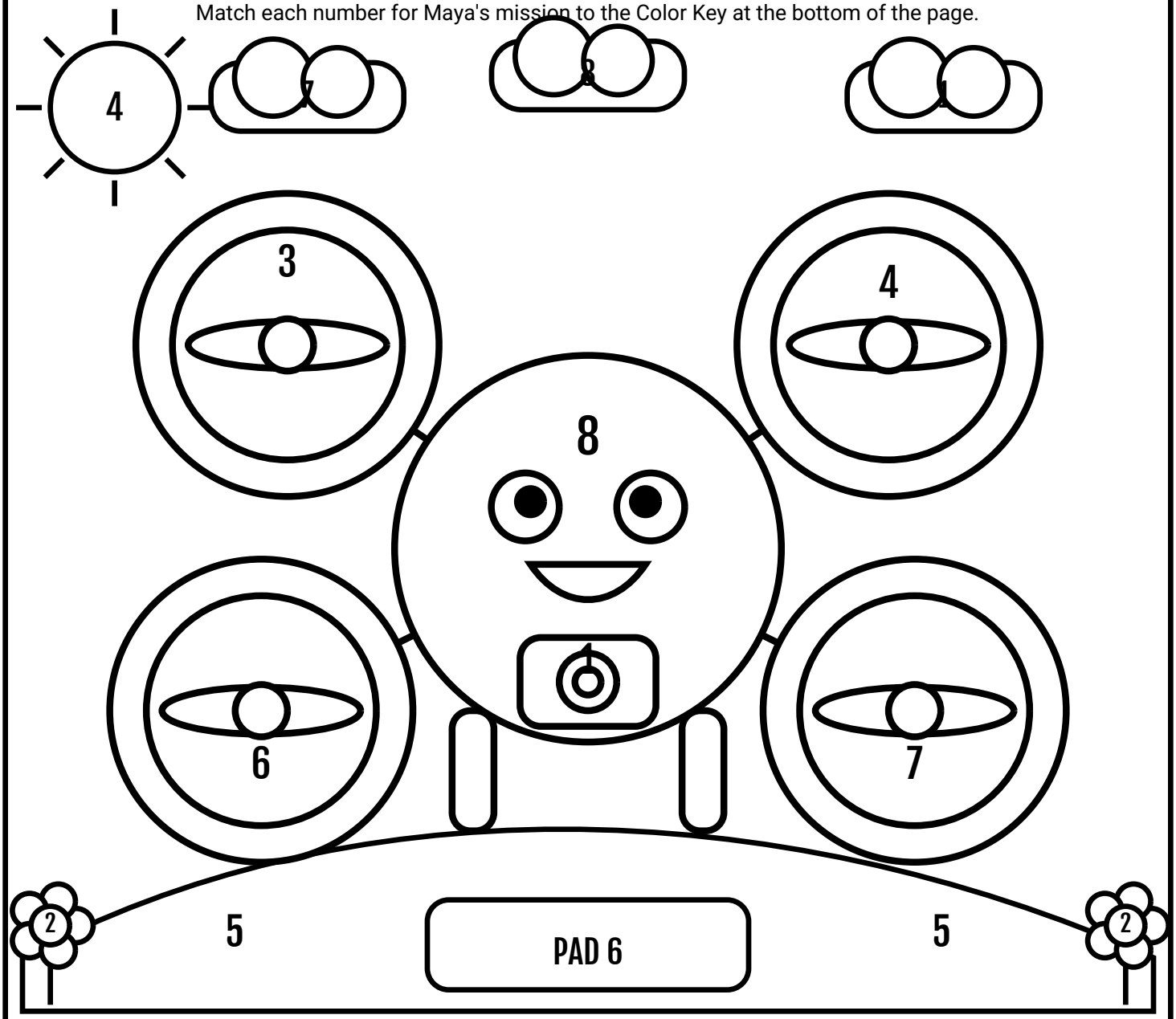


MAYA & PET DRONE – COUNT THE 4 PROPELLERS



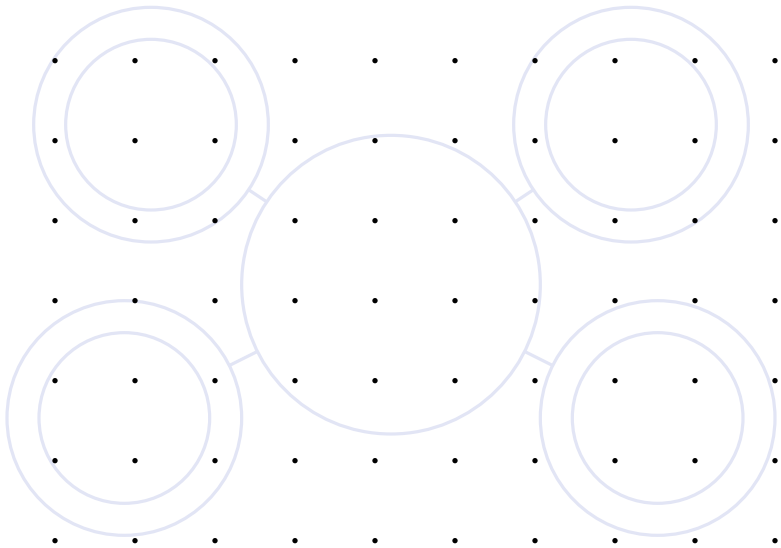
COLOR BY NUMBER: MAYA'S COUNT THE 4 PROPELLERS

Match each number for Maya's mission to the Color Key at the bottom of the page.



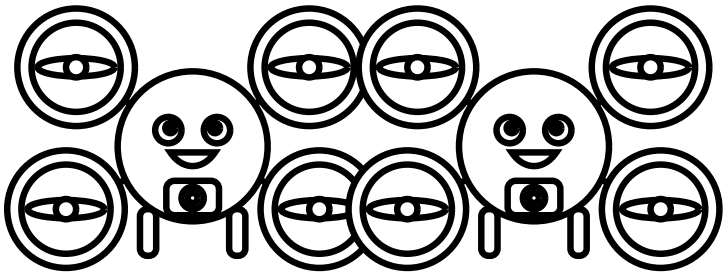
NUMBER-BY-NUMBER COLOR KEY

1 = RED	2 = BLUE	3 = YELLOW	4 = GREEN
5 = ORANGE	6 = PURPLE	7 = TEAL	8 = BROWN



DRONE MATH: MAYA'S COUNT THE 4 PROPELLERS

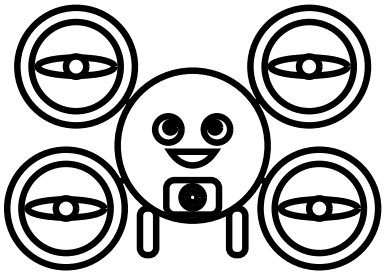
Every quadcopter has 4 ducted propellers! Solve Maya's flight problems and trace the numbers.



MISSION #1: COUNT ROTORS

2 Drones = 8 Props!

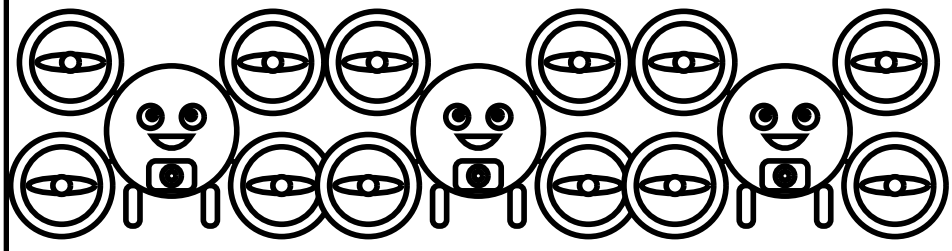
Color the drones & trace numbers!



MISSION #2: COUNT ROTORS

1 Drone = 4 Props!

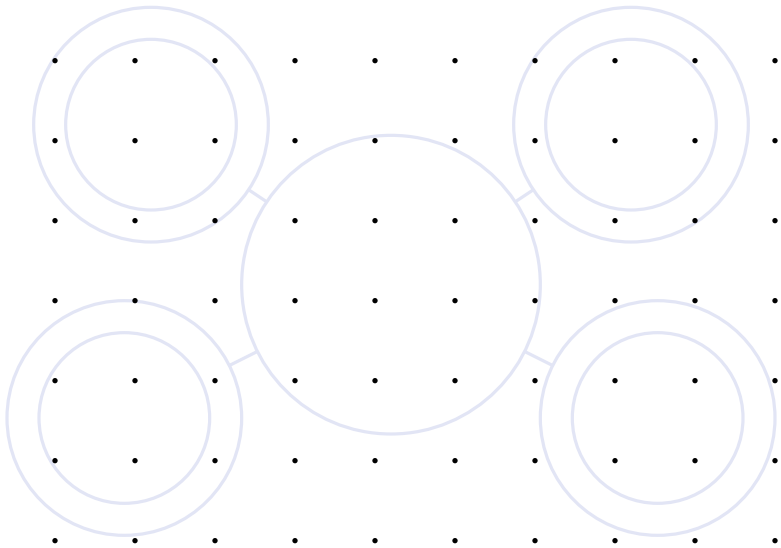
Color the drones & trace numbers!

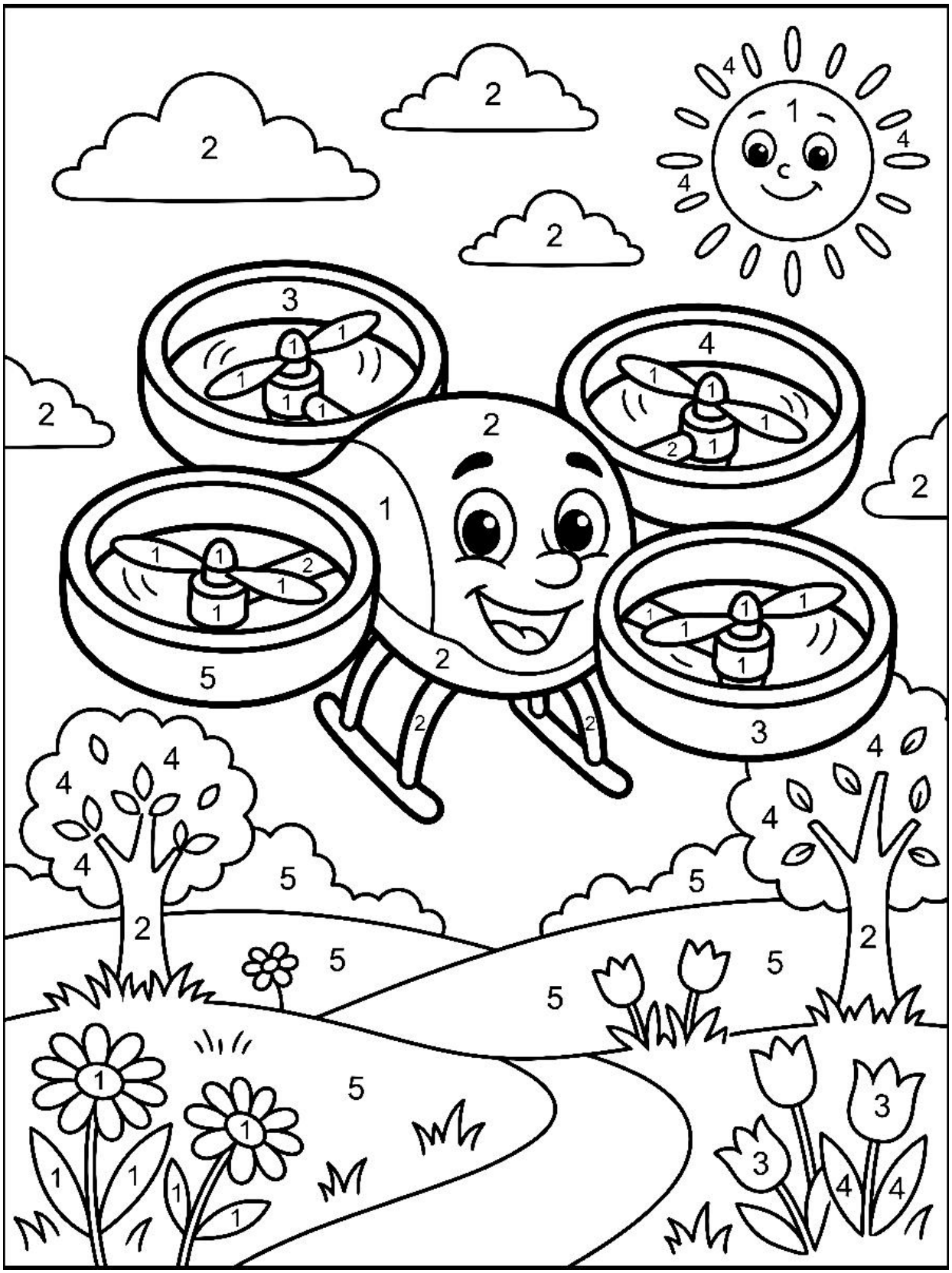


MISSION #3: COUNT ROTORS

3 Drones = 12 Props!

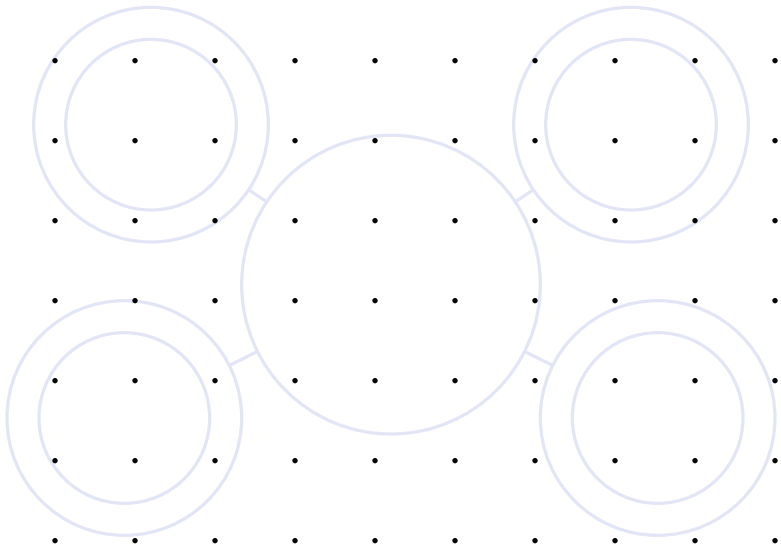
Color the drones & trace numbers!





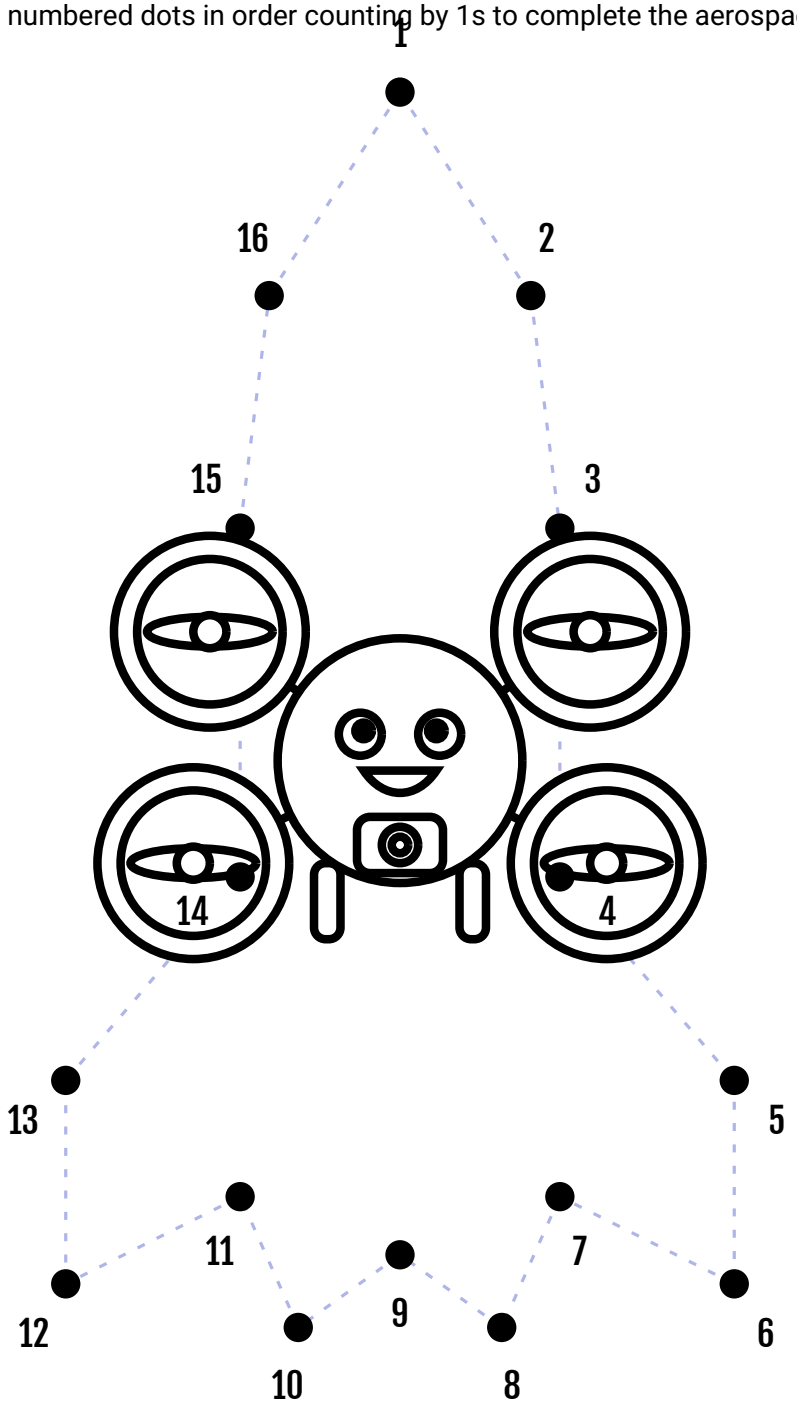
COLOR BY NUMBER & COUNT ROTORS WITH MAYA!

BLEED-PREVENTION BACKING PAGE 10 - BOOK #001 - FLIGHT LOG & SKETCH PAD



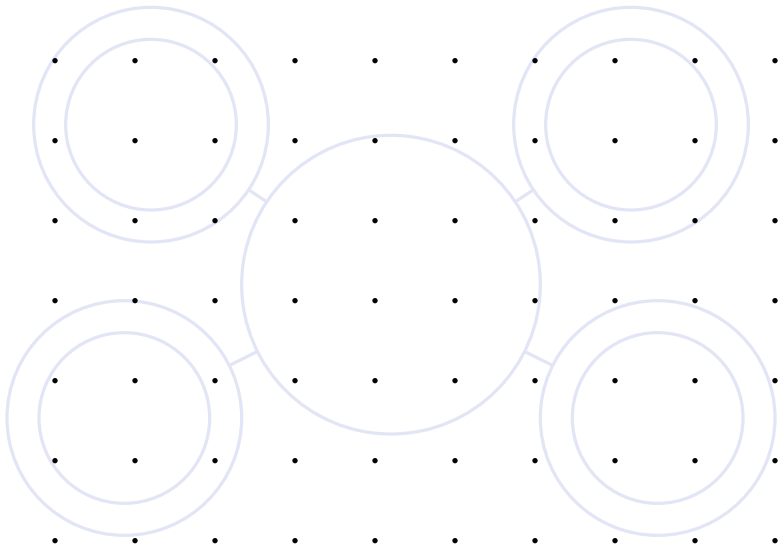
CONNECT THE DOTS: MAYA'S COUNT THE 4 PROPELLERS

Connect the numbered dots in order counting by 1s to complete the aerospace blueprint!



BONUS FOR MAYA: You connected 16 dots counting by 1s! Final number = 16!

BLEED-PREVENTION BACKING PAGE 12 - BOOK #001 - FLIGHT LOG & SKETCH PAD



CODE-BY-NUMBER: SOLVE & COLOR PIXEL DRONE

Solve or match the number in each grid square (A1-G7) using the Color Code Key below to reveal pixel art!

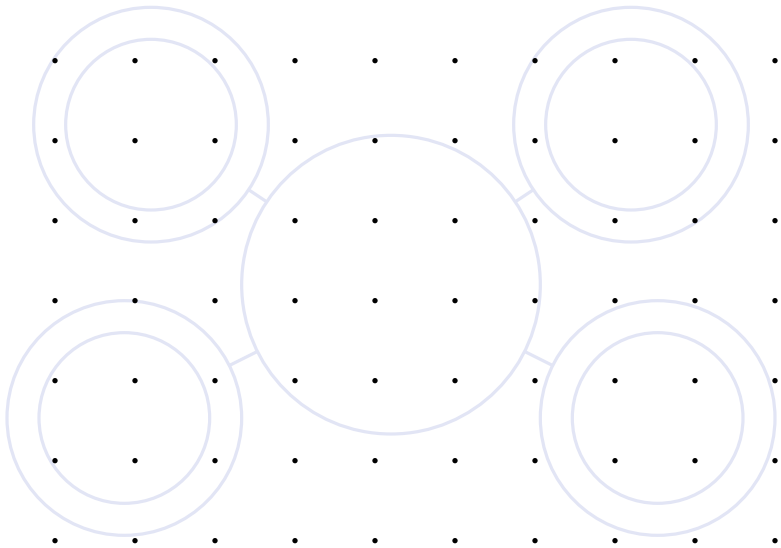
	1	2	3	4	5	6	7
A	5	5	5	1	5	5	5
B	5	5	1	3	1	5	5
C	5	5	2	2	2	5	5
D	5	5	2	3	2	5	5
E	5	1	2	2	2	1	5
F	1	1	3	1	3	1	1
G	4	4	4	4	4	4	4

PIXEL CODE COLOR KEY

1 = RED 2 = BLUE / INDIGO 3 = GOLD / YELLOW 4 = GREEN 5 = LIGHT SKY BLUE

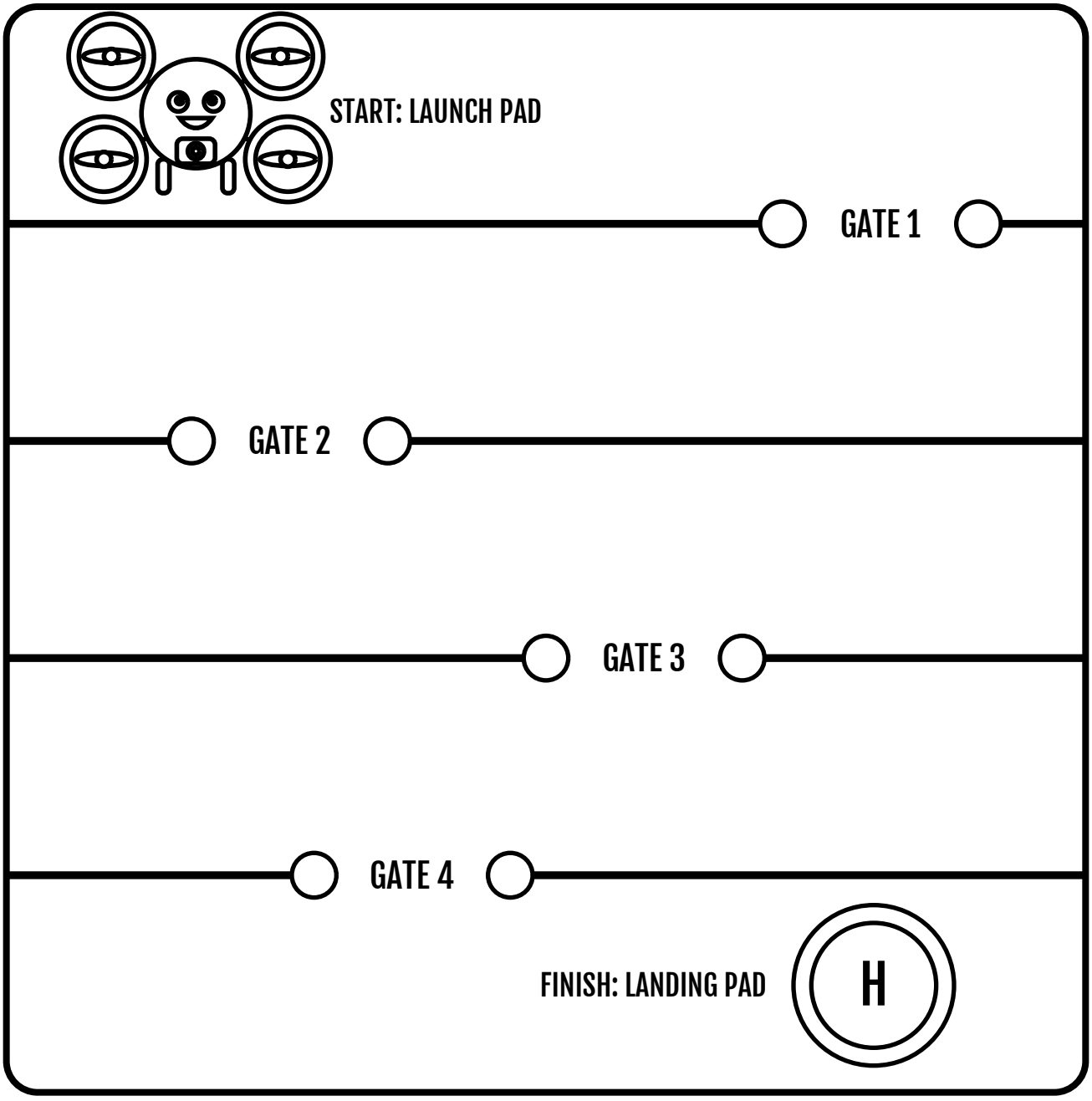
STEM Tip: Computers and drone cameras store pictures as grids of numbered pixels just like this!

BLEED-PREVENTION BACKING PAGE 14 - BOOK #001 - FLIGHT LOG & SKETCH PAD



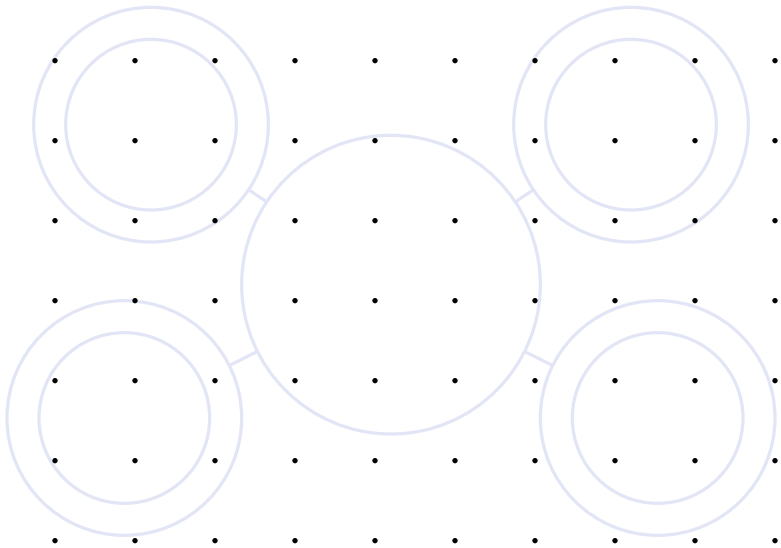
SAFE FLIGHT-PATH MAZE: GUIDE MAYA'S DRONE!

Draw a line from LAUNCH at top-left through open gates in order (1 -- 2 -- 3 -- 4) to LANDING PAD 'H'!



COLORING CHALLENGE: Color all 4 Gate Hoops different colors and draw arrows along your safe flight path!

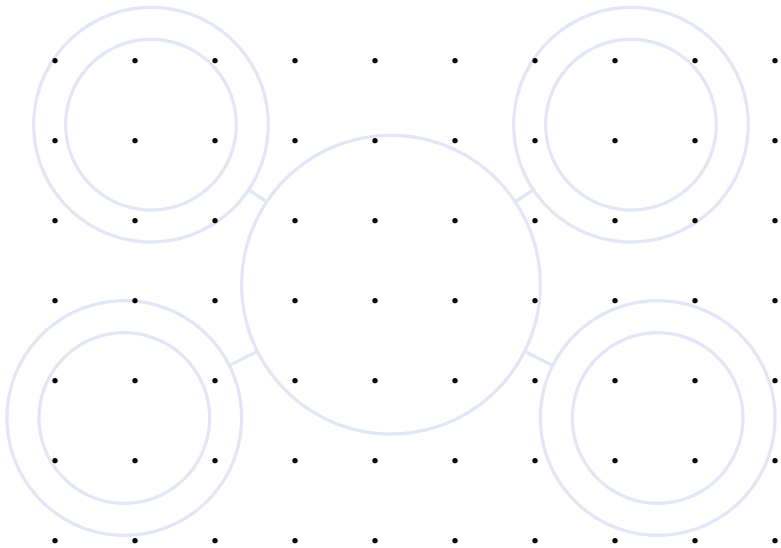
BLEED-PREVENTION BACKING PAGE 16 - BOOK #001 - FLIGHT LOG & SKETCH PAD





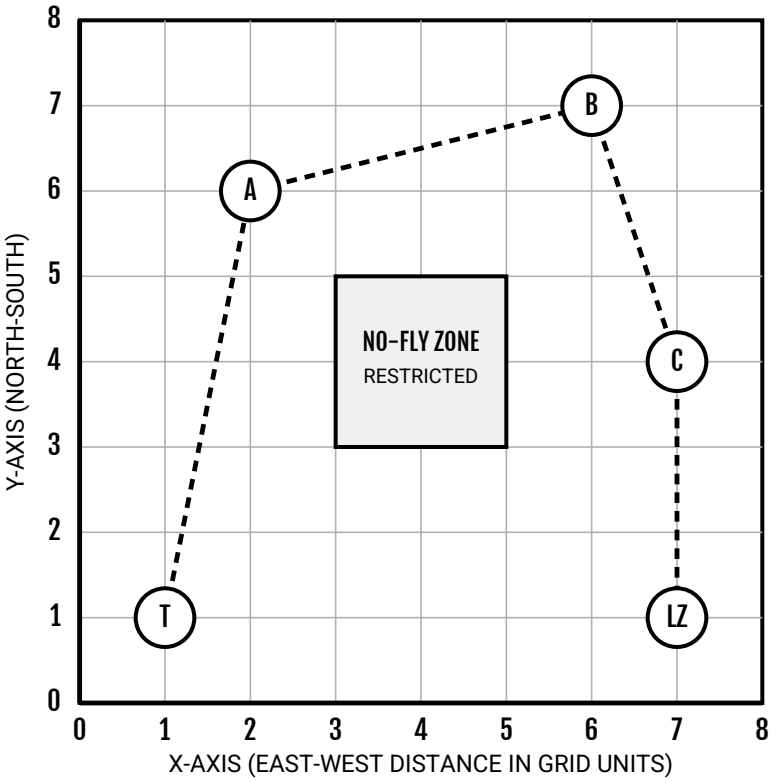
SAFE DRONE FLIGHT AT ORGANIC SUNFLOWER & CORN FARM

BLEED-PREVENTION BACKING PAGE 18 - BOOK #001 - FLIGHT LOG & SKETCH PAD



COORDINATE FLIGHT PLAN: MAYA'S WAYPOINT NAVIGATION

Plot waypoints on the (X, Y) grid for Maya's Count the 4 Propellers!



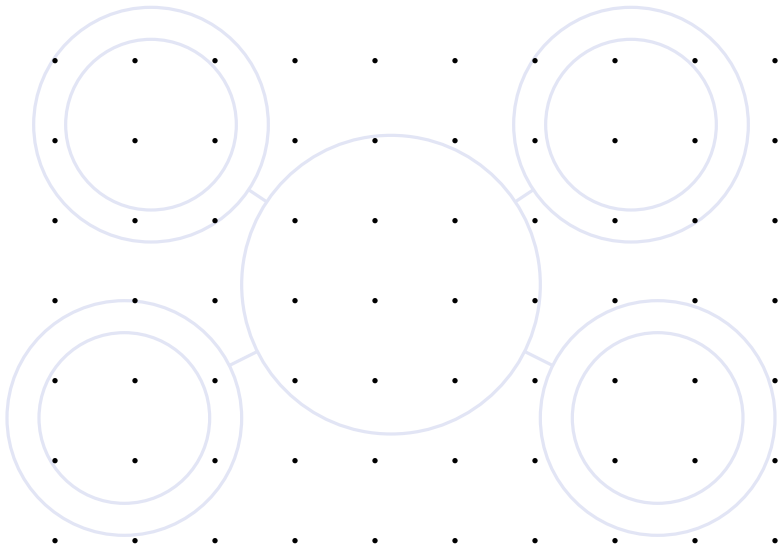
FLIGHT PLAN WAYPOINTS

WP T:	Takeoff Pad	(1 , 1)
WP A:	Maya Survey	(2 , 6)
WP B:	Waypoint Count the 4	(6 , 7)
WP C:	Bridge Check	(7 , 4)
WP LZ:	Landing Zone	(7 , 1)

COORDINATE CHALLENGE:

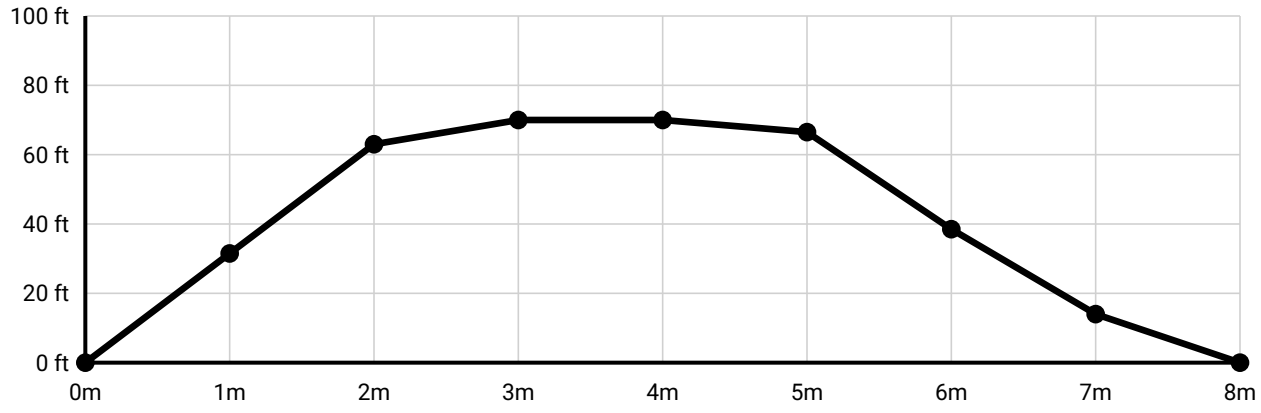
1. Waypoint B coordinates: (6 , 7)
2. Did flight avoid No-Fly Zone? [X] YES
3. Total flight legs flown: 4 legs

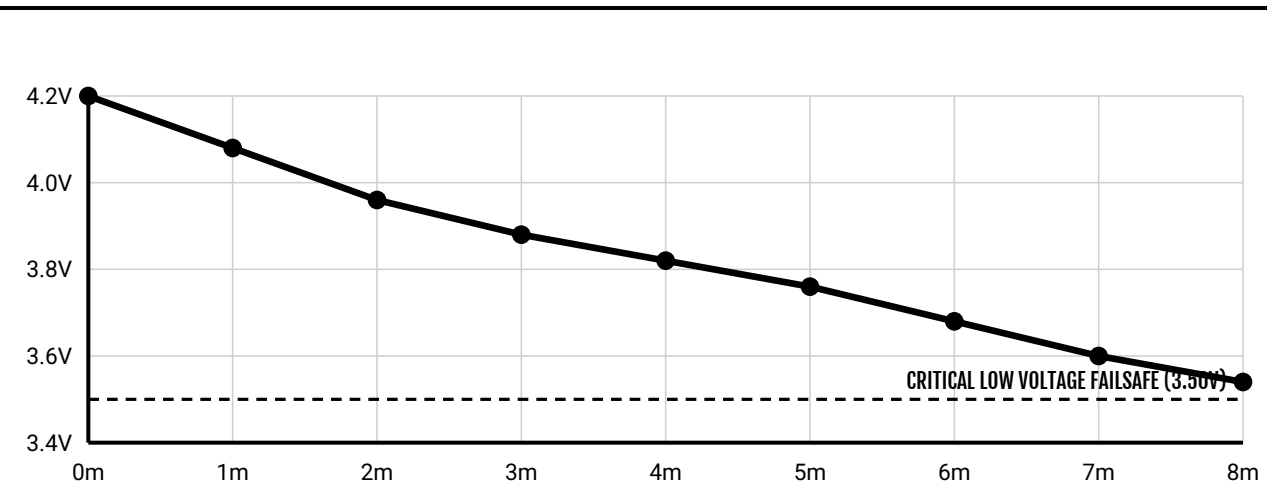
BLEED-PREVENTION BACKING PAGE 20 - BOOK #001 - FLIGHT LOG & SKETCH PAD



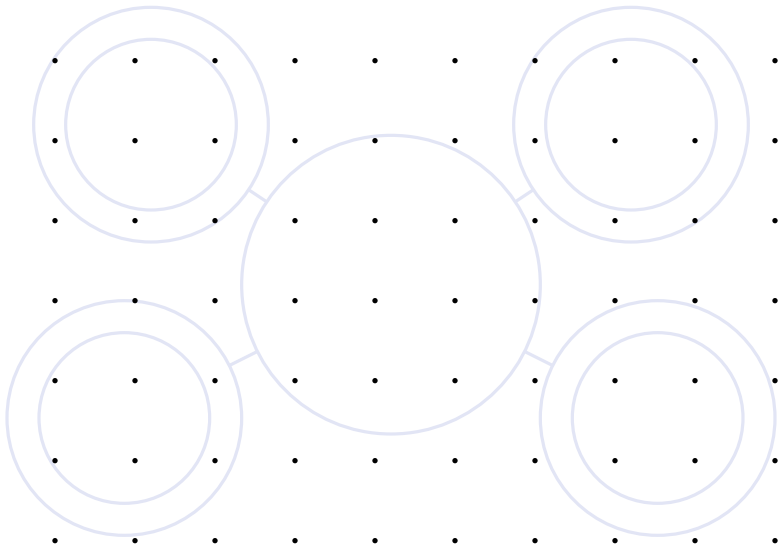
FLIGHT TELEMETRY: ALTITUDE & BATTERY DISCHARGE ENVELOPE

Analyze real flight telemetry data downloaded from the onboard BlackBox flight recorder.



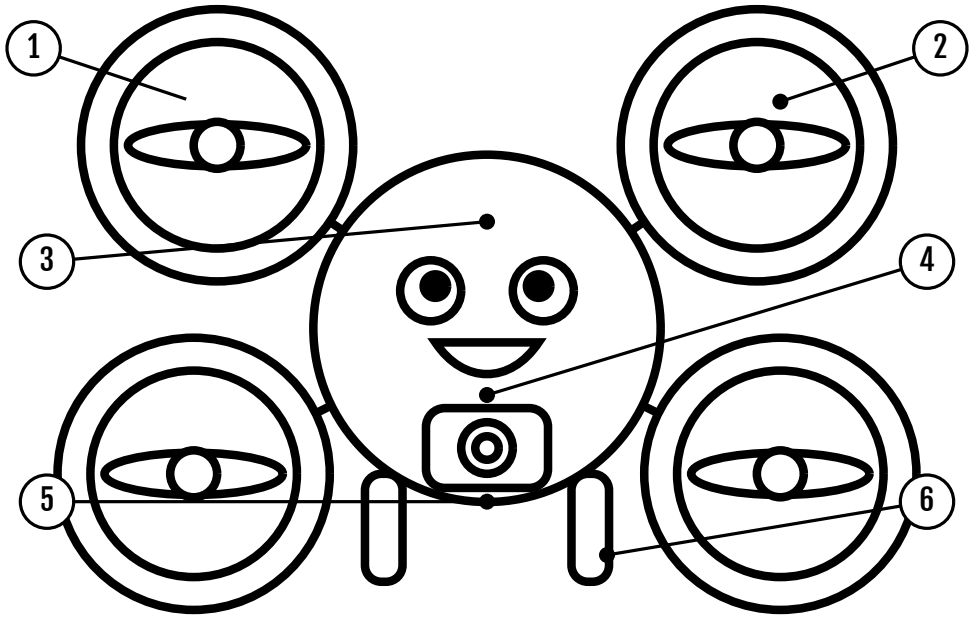


BLEED-PREVENTION BACKING PAGE 22 - BOOK #001 - FLIGHT LOG & SKETCH PAD



AEROSPACE BLUEPRINT: CUB-75 AVIONICS & AIRFRAME ANATOMY

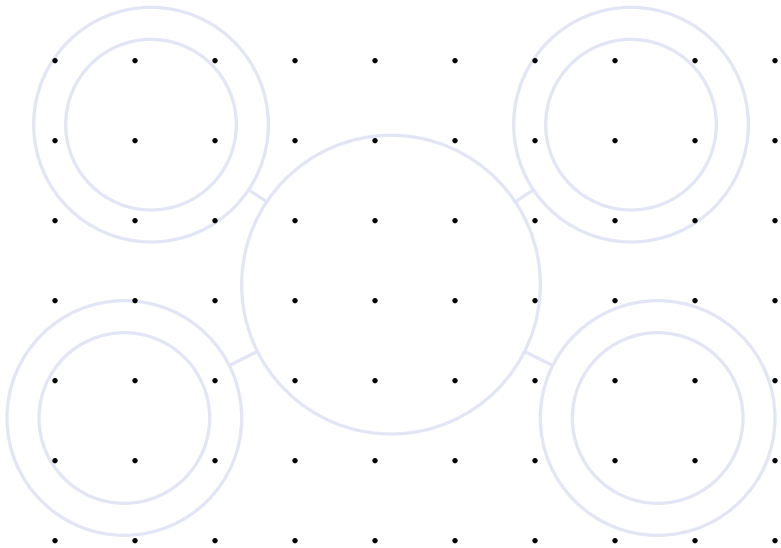
Study the mechanical airframe and avionics architecture of the Osito CUB-75 STEM training drone.



1	**Ducted Prop Guards**	Duct rings that shield propellers and provide 360-degree impact safety.
2	**Brushless DC Motors**	0802 19000KV motors that spin props at 30,000 RPM to produce vertical lift.
3	**Flight Controller (FC)**	Microprocessor & gyroscope that balances the drone 1,000 times every second.
4	**4-in-1 ESC Board**	Electronic Speed Controller regulating power sent from battery to each motor.
5	**Micro FPV Camera**	Ultra-wide lens sensor broadcasting real-time video to pilot FPV monitors.
6	**1S LiPo Battery**	3.8V High-Voltage Lithium Polymer rechargeable cell providing flight energy.

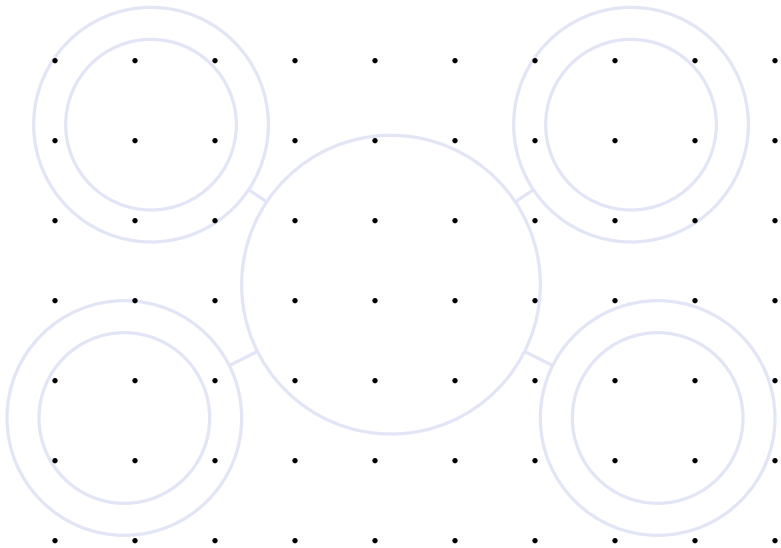
CUB-75 SPECS: Wheelbase: 75mm – AUW Weight: 39.5g – Motors: 0802 19000KV – Flight Time: 8.0 min

BLEED-PREVENTION BACKING PAGE 24 - BOOK #001 - FLIGHT LOG & SKETCH PAD



FLIGHT PHYSICS: THRUST-TO-WEIGHT & BATTERY ENDURANCE MATH

Apply real aeronautical engineering equations to calculate multirotor performance and endurance.



OFFICIAL OSITO JUNIOR PILOT FLIGHT MISSION LOG

RECORD YOUR ACTUAL FLIGHT MISSIONS AND COMPLETED STEM CHALLENGES

FLIGHT #	DATE	FLIGHT DURATION	MISSION OBJECTIVE	COACH SIGNATURE
Mission 1				
Mission 2				
Mission 3				

OSITO STEM PILOT CERTIFICATE OF ACHIEVEMENT

THIS CERTIFIES THAT JUNIOR PILOT

PILOT NAME HERE

HAS COMPLETED ALL STEM ACTIVITIES AND SAFETY PROTOCOLS IN

BOOK #001: COUNT THE 4 PROPELLERS

Core Aeronautical Skill: Counting by 4s & Rotor Symmetry

AND IS OFFICIALLY RECOGNIZED AS A CERTIFIED JUNIOR OSITO DRONE PILOT!

JUNIOR PILOT SIGNATURE

COACH / PARENT SIGNATURE & DATE

BLEED-PREVENTION BACKING PAGE 28 - BOOK #001 - FLIGHT LOG & SKETCH PAD

